

PROJECT PROPOSAL

Date of proposal:
Project Title: A RAG Based Multi Agent System for PC Build and Laptop Recommendations in Singapore
Group ID (As Enrolled in Canvas Class Groups): Group Members (name , Student ID):
Sponsor/Client: <i>(Company Name, Address and Contact Name, Email, if any)</i>
Background/Aims/Objectives: Choosing a computer requires users to compare configurations, local prices and user feedback while balancing budget and intended use. Desktop buyers also need to check component compatibility, while laptop buyers must consider configuration differences and portability. This project aims to develop a conversational recommendation system for users purchasing computer components or laptops in Singapore. It will interpret user requirements, retrieve relevant product information and reviews, recommend suitable options with traceable sources, and support follow-up adjustments. The objectives are to improve transparency in computer selection and evaluate whether specialised agents improve recommendation quality compared with a single-agent baseline.
Project Descriptions: Scope and user interaction. The proposed web application will accept requirements such as budget in SGD, intended software or games, preferred device type and existing components. It will ask for essential missing information, present desktop build lists or specific laptop configurations, explain trade-offs, and allow users to compare options or request changes while retaining selected constraints. Data foundation. Initial collection on 6 September 2026 produced 2,236 product-variant price records from Dynacore Singapore and Vii PC Singapore, together with 15 user-review excerpts obtained through public web retrieval. Records retain source links and collection timestamps. These review excerpts are a small sample; reviewer location and exact configuration matches are not always verified. Additional product specifications and review coverage are needed. RAG pipeline. Product records will be cleaned and kept separate by configuration. Long descriptions will be divided into meaningful passages, while short reviews will remain intact. Embeddings and source metadata will be stored in a vector database for retrieval. Current prices and stock will remain available through structured queries. Retrieved evidence will support explanations, with insufficient evidence identified explicitly.

Multi-agent collaboration. A coordinating agent will extract requirements and route tasks to a desktop-planning agent or laptop-selection agent. Both may run when the device type is undecided. An evidence agent will retrieve relevant feedback, and a review agent will check requirement coverage and supporting sources. Revisable issues will return to the relevant planner within a bounded correction loop.

Backend and validation. FastAPI will expose the application services, and LangChain will support model and retrieval interactions. Programmatic tools will calculate totals, filter stock and check compatibility against available specifications. Missing specifications will produce an unknown status rather than a compatibility pass. The interface will show configuration details, prices, source dates and unresolved limitations.

Evaluation and deliverables. The project will deliver a data preparation and indexing pipeline, a RAG-based multi-agent backend and a web interface for recommendations and adjustments. Evaluation will compare single-agent and multi-agent approaches using the same data and test requirements, measuring constraint satisfaction, source accuracy, known compatibility-conflict detection, adjustment success, response time and model usage cost.